

THERMAL ELECTRIC GENERATOR

INTRODUCTION :

When generating electricity in power stations, around two thirds of the energy is lost in the form of waste heat that is discharged from cooling towers. The main reason is that the gas or steam-powered turbine systems, that operate to produce most of the electrical power, primarily function by burning fuel to produce energy in the form of heat. This is followed by the conversion of this heat energy into mechanical energy within the turbine, and finally turning the mechanical energy into electrical energy in a generator. As a result, only about 1/3 of the energy released from the fuel actually ends up in the transmission lines leaving the power plant. The ability to collect the heat wasted within these processes and convert it to usable electrical power would enormously increase the efficiency of power generation. Additionally, the reduction of greenhouse emission from the reduced wastage would be beneficial for the environment as less fuel is burned for the same amount of electricity produced.

In recent years thermoelectric generator (TEG) systems have attracted great consideration in the recovery of waste heat due to their incomparable Advantages :
TEGs provide an opportunity to generate electrical energy from heat energy without the need for moving

parts such as turbines, which eliminates extra costs resulting from maintenance and replacement . TEGs have no economy-scale-of effect and can be utilized for micro generation in a restricted position or can be used to generate kilowatts. TEGs are also environmentally favourable as they operate with no sound pollution. Conversely, TEGs do have a low energy conversion efficiency and require a relatively constant heat source, which are disadvantages.

WORKING PRINCIPLE :

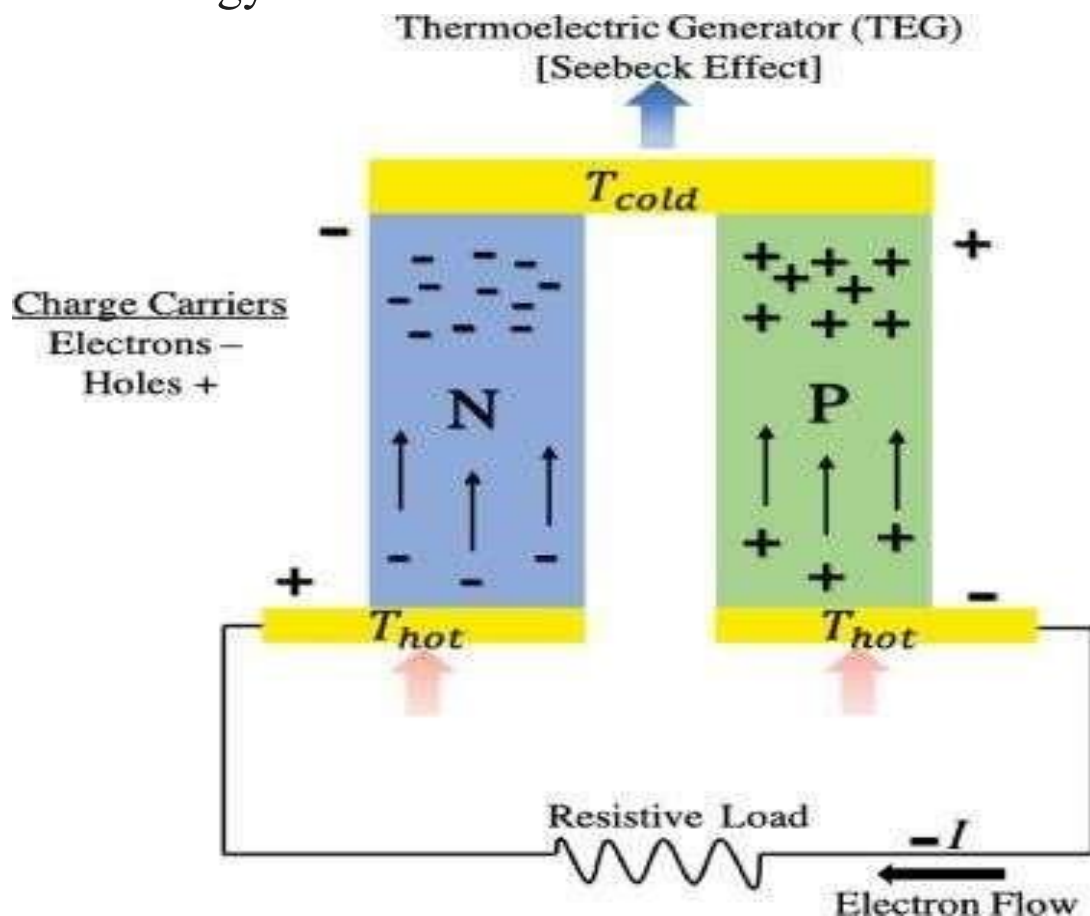
TEG-working principle

Mostly, TEG systems consist of three key elements:

- .1) A heat exchanger (HEX); This absorbs the heat and transfers it into the thermoelectric modules.
- .2) Thermoelectric modules (TEMs); The TEMs generate electricity when a temperature difference exists between their ends. A TEM contains many pairs of thermoelectric couples, and each couple normally combines a pair of p- and n-type semiconductors.
- .3) A heat sink; In order to dissipate the additional heat from the thermoelectric modules.

The temperature difference between two sides of the generator is what determines the operation of TEGs. The below figure explains the theory behind TEGs. If one side of a piece of metal could be heated while

simultaneously cooling the other side, electrons surrounding the metal atoms at the hot side will have more energy than the equivalent electrons at the cooler side, this means the hot electrons will have more kinetic energy than those in the cooler side



A thermoelectric generator works on the principle of thermoelectric effect. Thermoelectric effects mean, a temperature gradient is created on the two sides of the generator. The temperature gradient means the difference in temperature levels. For example, if we take a metal rod, and heat one end, and cool the other end. A temperature gradient is created between the two ends.

The electrons present at the hot end would be at a high energy level as compared to electrons present at the cool end side. This means that the hot electrons will try to tend to move towards the cool end due to the temperature gradient.

Due to the movement of electrons, the hot end will tend to be positively charged, and the cool end is negatively charged. Since the electrons at both ends are at different energy levels, they tend to move towards each side, relatively at different speeds. The hot end side electrons would move faster and create a potential difference between the two ends. The creation of potential difference due to temperature gradient, at the two points, is defined as the thermoelectric effect. The same principle is used in thermoelectric generators.

SEE BECK EFFECT:-

This mechanism, where a temperature differential produces a voltage, is known as the Thermoelectric Effect or Seebeck Effect .

The Seebeck effect explains the build-up of a potential gap (ΔV) through a semiconductor (or conductor) because of the diffusion of its load carriers over a temperature gradient ($\Delta T = T_{\text{hot}} - T_{\text{cold}}$). It is encountered by the substance if one side of it is heated or cooled.

The Seebeck Coefficient sign shows that the material is a p-type or n-type conductor, the positive sign indicates the p-type, the negative indicates the n-type material. The Seebeck coefficient (S), which is responsible for allowing the current to move while the temperature gradient is present, is the relation between the theoretical differential created and the temperature gradient applied which may be seen Below:

$$S = \frac{\Delta V}{\Delta T}$$

CHOPPER CIRCUITS

Definition: The circuits that are designed to directly convert fixed dc voltage into adjustable dc voltage are called Chopper Circuits. It is also known as dc to dc converter or dc chopper. Choppers are considered to be dc equivalent to ac transformers, the reason for the same is that similar to the transformers, choppers hold the ability to step up or step down the fixed dc input voltage.

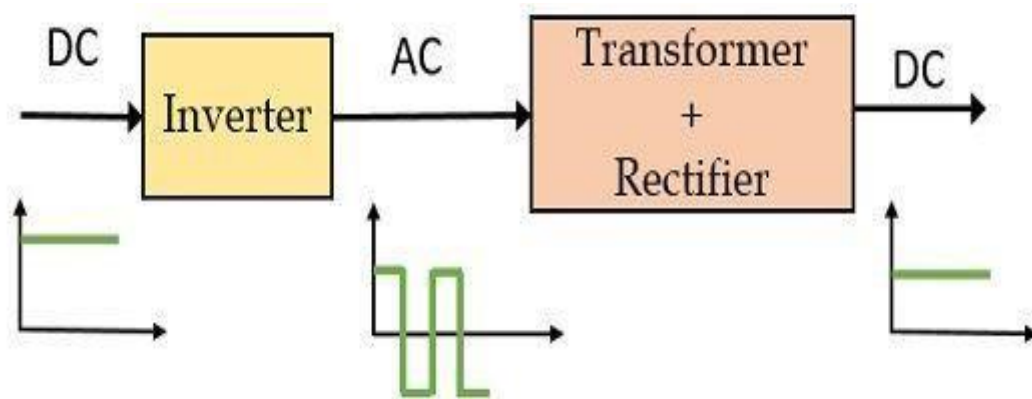
why do we need dc to dc conversion?

We are aware of the fact that a large number of applications need power from dc voltage sources and the operation goes quite well when the input provided is from the variable dc voltage source. For example, we know that the household power supply system works on 240 V ac supply i.e., input is of fixed nature. Thus, in order to power the devices that need dc input and that too of variable nature, power conversion becomes necessary.

Basically, chopper circuits are designed to provide adjusted dc power according to the requirement. In order to convert fixed dc voltage into adjustable one, two types of dc to dc conversion techniques can be implemented. The converters are as follows:

1. **AC Link Chopper:** In this type of chopper circuit, first the fixed dc signal is converted into an ac signal

using an inverter. Further, the obtained ac signal is provided to a transformer that performs the step-up or step-down operation. The output of the transformer is provided to a diode rectification unit that provides a dc signal as output.

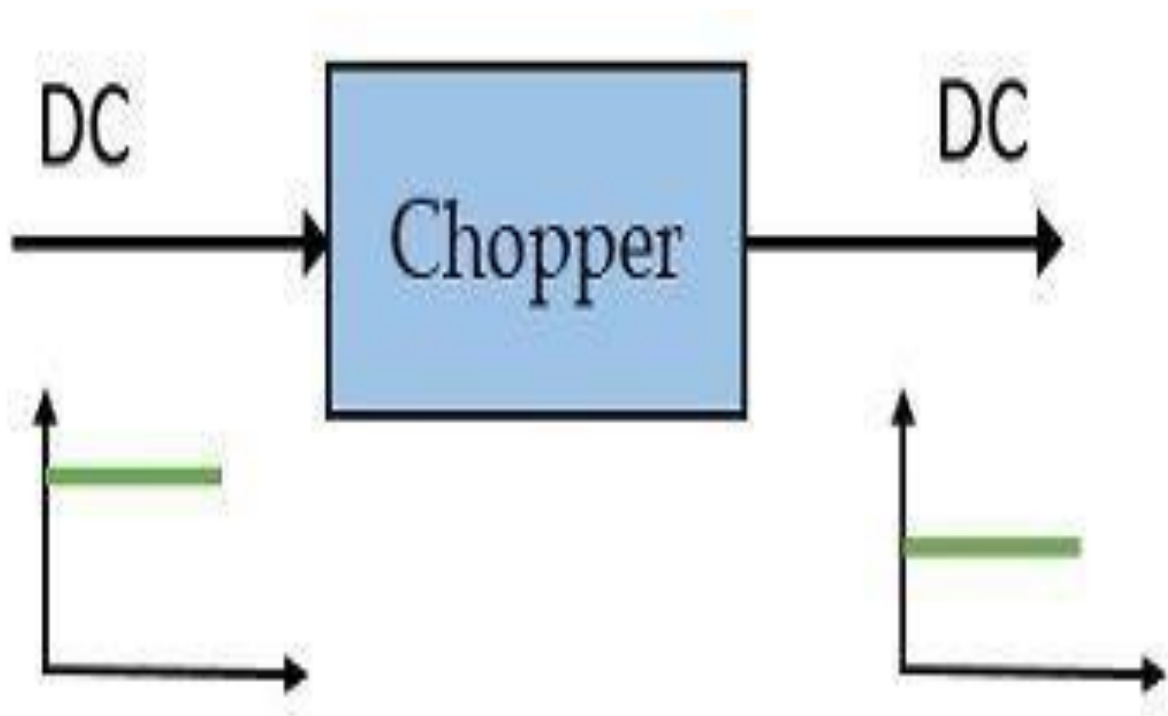


It is a two-stage conversion process as at first dc is converted into ac and then ac is changed into dc. This two-stage conversion process is less efficient and is of high cost.

- 2. DC Chopper:** This type of chopper circuit is considered to be static that changes fixed dc input voltage into a variable one. It behaves in a similar manner as that of an ac transformer and executes single-stage conversion in order to convert dc into dc. This conversion technique is more efficient than the one involved in ac link chopper.

The various power semiconductor devices that are used in chopper circuits are power BJT, power MOSFET, GTO, or IGTB. The representation of these devices in any circuit is generally represented by a switch with an arrow. In the open condition of the switch, no flow of current takes place.

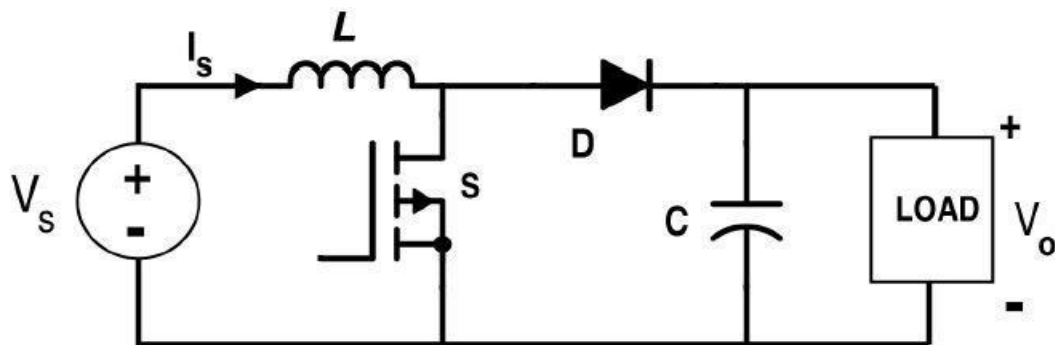
But as soon as the switch is closed, the current begins to flow in the direction of the arrowhead.

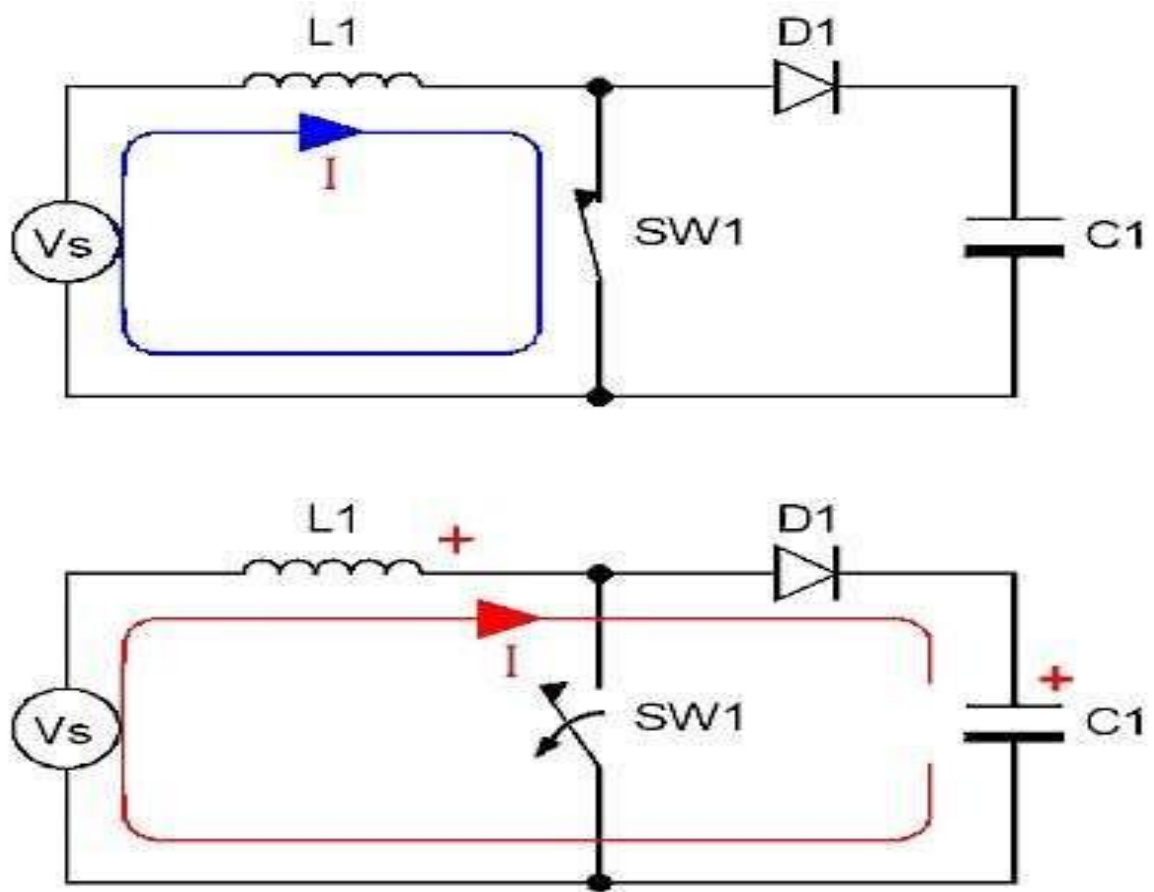


Design of boost converter

- Basically we are implementing a voltage booster which can be useful in day to day life.
- For amplifying a certain voltage, basically we start designing circuit by using switches.
- Here, we are using two switches let's name them as lower switch and higher switch.
- Here, we assume lower switch as a P-channel MOSFET because voltage drop across MOSFET when on, is less than that for BJT(since BJT is also used as a switch). Also it is expected to reduce power loss at DC – DC converter(since it has less R_{ds}).

Circuit Diagram



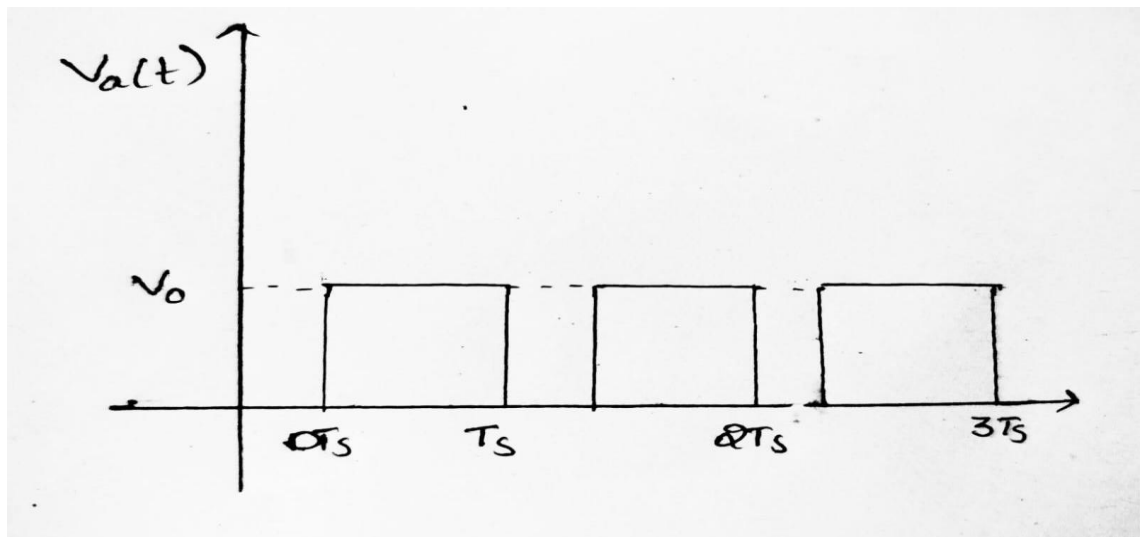


Working :

- As already mentioned, MOSFET and diode are lower side and higher side switches.
- Here, we use Schottky diode because it has less voltage drop in forward bias (makes circuit more efficient).
- Firstly, we close the switch on lower side and open the upper side switch.

- Then, the current flows through the inductor and build up a magnetic field across it.
- If we open the lower side switch and close the upper side switch, still the same current flows through the inductor since inductor does not allow sudden current changes and this current will be registered as output.

AT THE NODE V_a :-



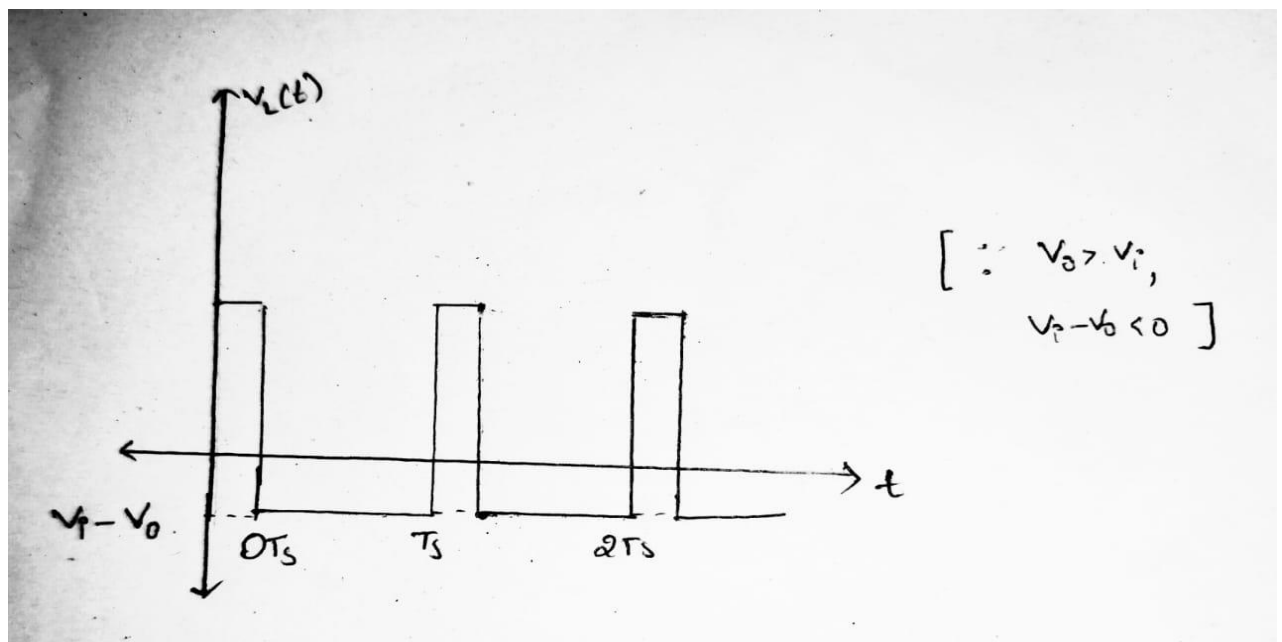
Here, both MOSFET and diode work simultaneously.

Let T_s be the switching period. Let D denote duty cycle then the

MOSFET works as a closed switch for the time DT_s . After this time, MOSFET becomes

open and diode acts as closed switch. So, from the above graph we can observe that in the time between 0 to DT_s the node V_a is shorted to ground. So, the output is zero. After the time DT_s current flows through diode and the output will be V_o .

AT THE INDUCTOR:



At the inductor, when switch 1 i.e MOSFET is closed then input source and inductor are in series so output will be V_o itself. Then after time DT_s we apply KCL there and we see that voltage across inductor will be :-

- $V_L(t) = V_i - V_o$ since our converter gives gain in voltage i.e $V_o > V_i$. $V_L(t) < 0$. We know that

$$I_L = \frac{1}{L} \int_0^{DT_s} V(T) dT$$

$$I_{\max} = \frac{1}{L} DT_s V_i + I_{\min} \quad (1)$$

$$I_{\min} = \frac{1}{L} \frac{(V_i - V_o)(1-D)Ts}{L} + I_{\max} \quad (2)$$

On solving equations 1 & 2

We obtain the result:-

$$V_o = \frac{V_i}{1-D} \quad (3)$$

Here we assume the circuit is working in continuous conduction mode.

Here we apply conservation of power i.e power remains same

$$P_{in} = P_{out}$$

$$V_i I_i = V_o I_o$$

$$I_o = (1-D) I_i$$

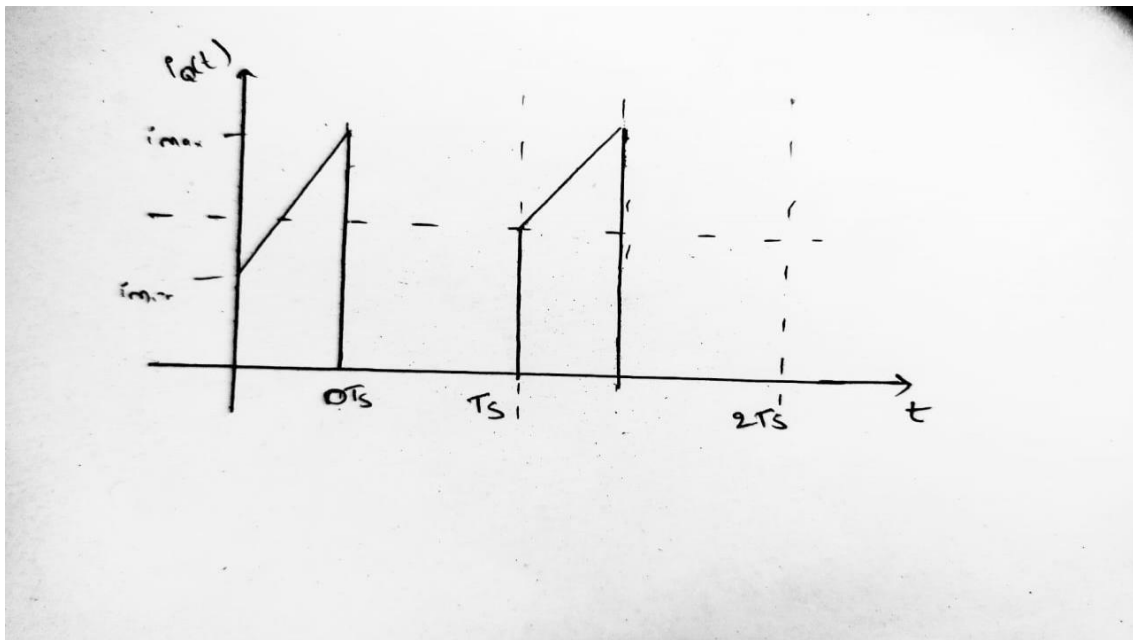
$\Delta I/L = I_{\max} - I_{\min}$ From eq 1 we obtain we can write that

$$L = \frac{V_i D}{f_s \Delta I_l}$$

On substituting V_i from eq3, also $V_o = R_L I_o$

$$L = \frac{R_L (1-D) D I_o}{f_s \Delta I_l}$$

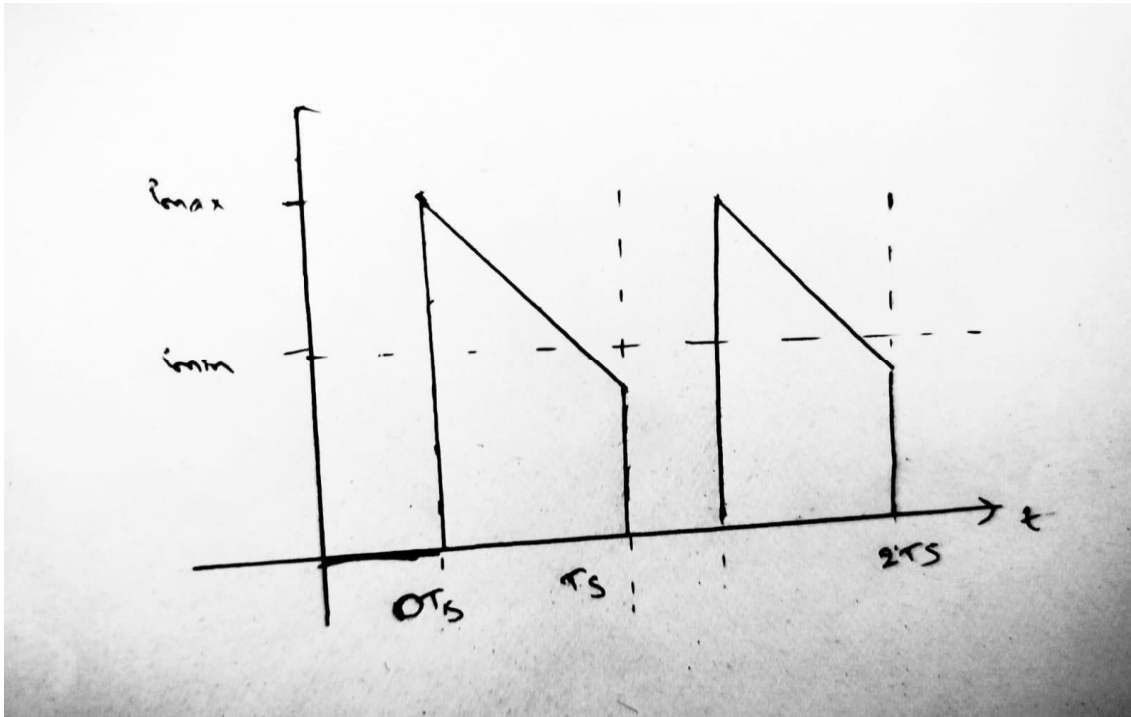
Now we study the behaviour of MOSFET,



For working of the mosfet ,its $V_{ds} > V_i$ Avg Current through mosfet

$$= \frac{I_o^2 R_{ds} D}{(1-D)^2}, \text{ where } R_{ds} \text{ is the drain to source resistance.}$$

In diode:-



$V_{\text{reverse for diode}} = V_0$

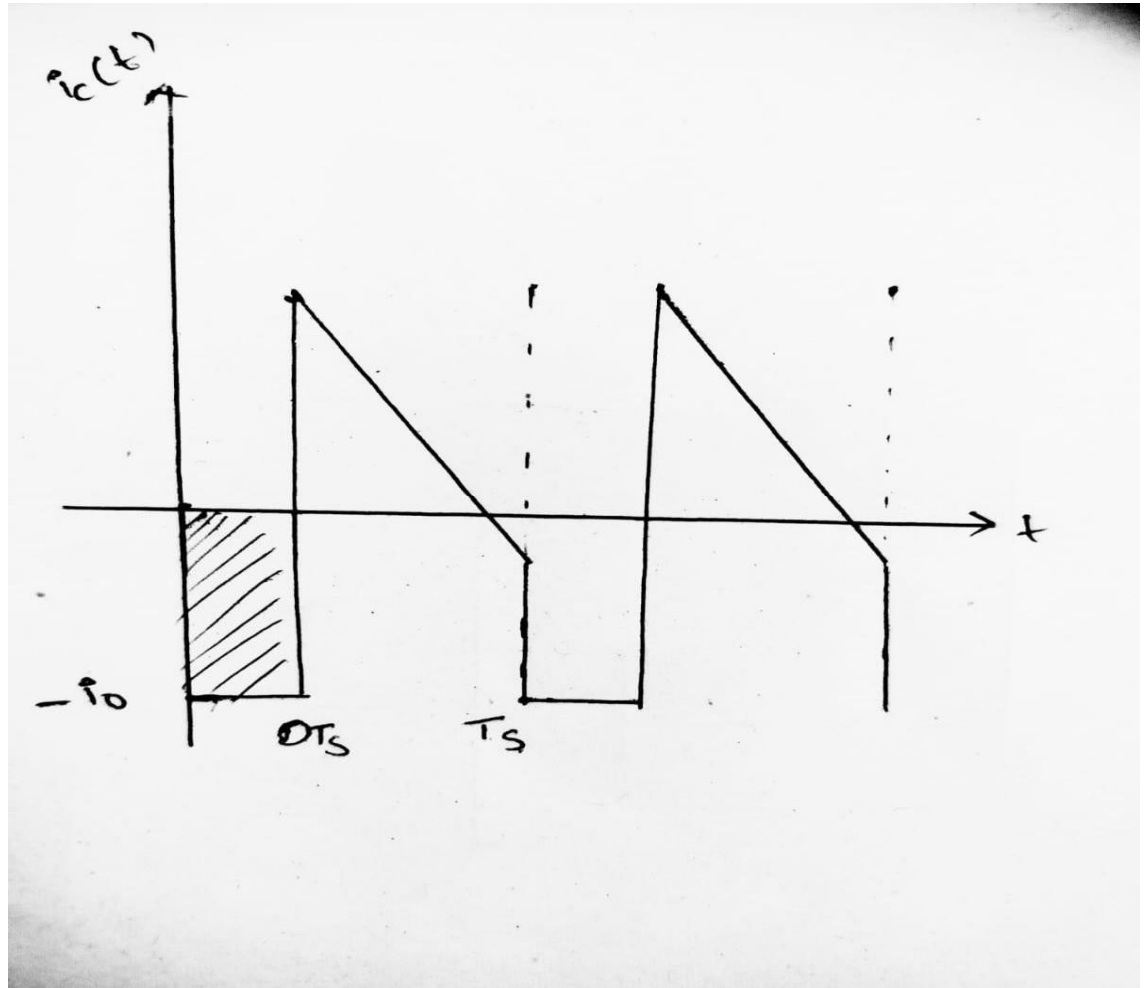
$$\text{current through diode} = \frac{I_o}{1-D}$$

avg current through it = current
through it * (1-D)
= I_o

Power dissipated = $I_o V_{\text{forward}}$ (to
minimize this we use Schottky diode)

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In capacitor:-



In the time where diode is acting as open switch, capacitor has to supply current to load. so the direction of current is opposite i.e we denoted I_c in opposite sense.

Charge initially $\Rightarrow$ area under i - t graph $= I_o DT_s$

$$\Delta V = \frac{I_o DT_s}{C} \quad C = \frac{I_o D}{\Delta V f_s}$$

- Here we prefer electrolytic or multilayer ceramic type
- To minimize the equivalent source resistance to reduce loss

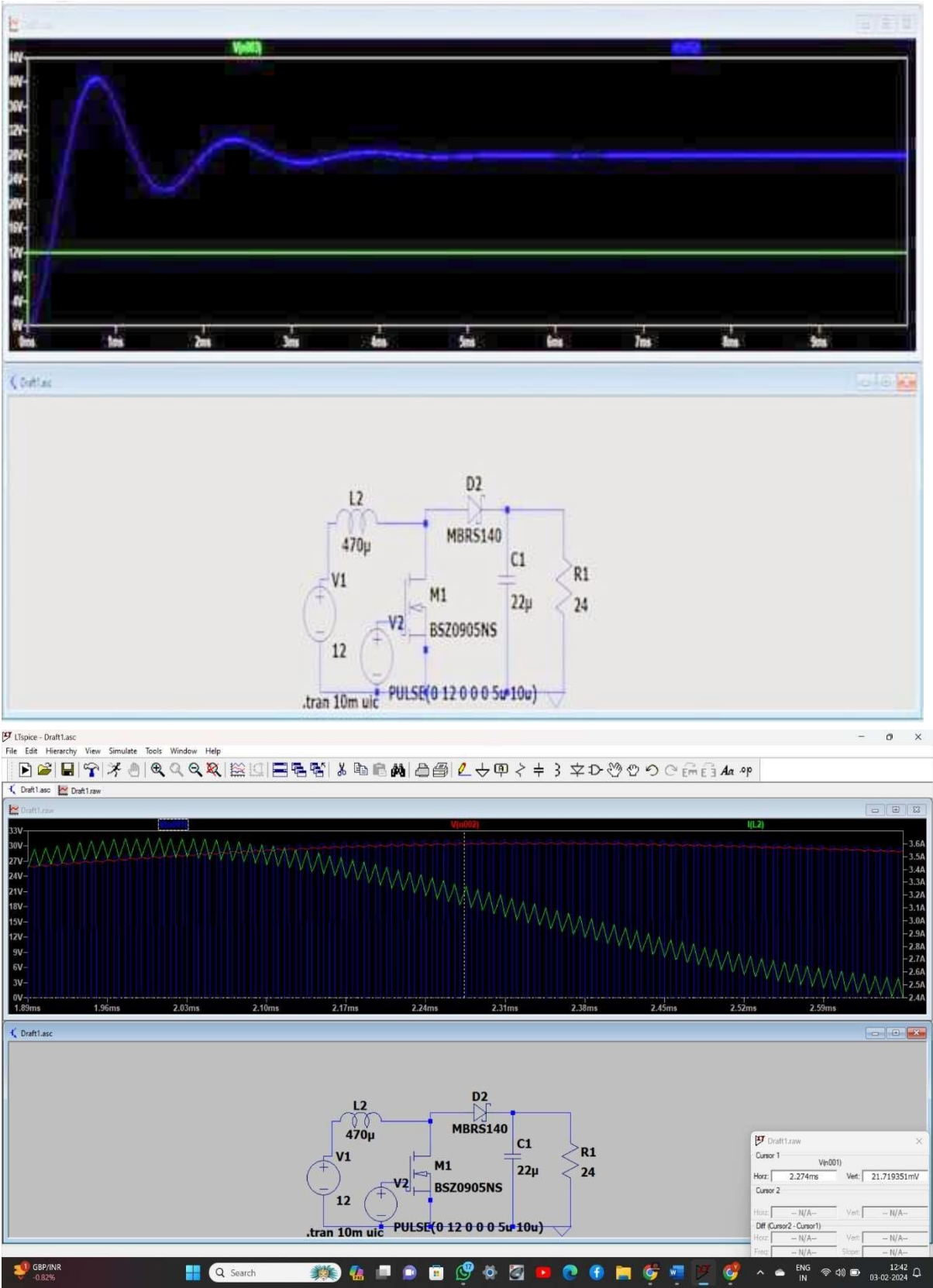
Now we relate duty cycle with efficiency of converter

Where η =efficiency of converter , e.g. estimated 80% The efficiency is added to the duty cycle calculation, because the converter has to deliver also the energy dissipated. This calculation gives a more realistic duty cycle than just the equation without the efficiency factor.

Method to increase efficiency:-

The parallel connection of diodes and MOSFET switches in the boost converter circuit is good way to increase the efficiency of this converter. Also, it makes the switches of the converter share the same thermal and conduction losses and thus, it can increase the output power of the converter. The individual MOSFET gate resistances that are fired by a proper driver circuit are better in efficiency than the common gate resistance connection

LT SPICE SIMULATION:



- Graph 1 indicates the output and input voltages for the implemented demo circuit.
- The above circuit is designed to obtain the output of around 28V ,by assuming it has a 100% efficiency.
- The second shows the ripple current values which matched the above discussed theoretical graphs.

References:

- 1) Robert W. Erickson: Fundamentals of Power Electronics, Kluwer Academic Publishers, 1997
- 2) Understanding Boost Power Stages in Switch mode Power Supplies (SLVA061).
- 3) Video lectures by DR K ,power electronics.

Conclusion:

In Conclusion, the boost converter experiment yielded, the valuable insights into the performance characteristics of the circuits under varying input conditions. The analysis of voltage wave forms and efficiency measurement

demonstrated the converter's ability to efficiently step up the input to the desired output levels.

Furthermore, the derived equations and graphical representations provided a comprehensive understanding of the converter's behaviour in the continuous conduction mode. The choice of MOSFET and diode components proved effective, contributing to the overall success of boost converter design. This experiment not only enhanced our practical understanding of power electronics but also underscored the importance of component selection and circuit design in achieving optimal performance.

Result :-

Finally we have implemented our boost converter and expected output is obtained .we can vary input voltage for obtaining different outputs.